



**WEAPONS
ROBOTICS AND
CONTROL
ENGINEERING**
UNITED STATES NAVAL ACADEMY

HOPPER INFORMATION SERIES

02 / FORCES AND MOTION

How a quadrotor moves

From four rotor thrusts to vertical, linear and angular motion.

THRUST

PITCH + ROLL + YAW

SIMPLE MODEL



Total thrust sets vertical acceleration

TOTAL THRUST

$$T = \sum_{i=1}^4 T_i = T_1 + T_2 + T_3 + T_4$$

HOVER CONDITION

$$T = mg \quad \Rightarrow \quad \ddot{z} = 0$$

Read the physics

- Every rotor contributes an upward thrust magnitude.
- Hover requires total thrust to balance weight.
- More total thrust accelerates up; less accelerates down.

FIRST-ORDER PROPELLER MODEL

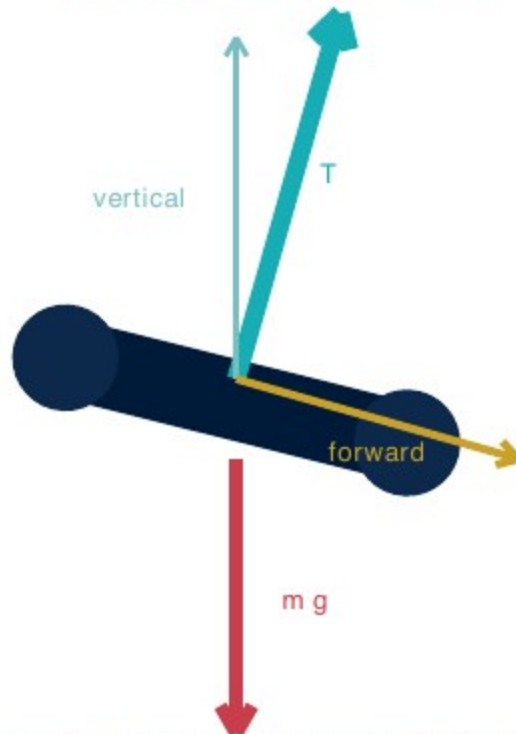
$$T_i \approx k_f \omega_i^2$$

The square-law model is useful for intuition; the onboard controller handles the real motor/propeller details.



Translation comes from tilting the thrust vector

SIDE VIEW: PITCH FORWARD



The horizontal part of thrust accelerates the aircraft.

PITCH

Tilt nose forward/backward. Total thrust gains a forward/backward component.

ROLL

Tilt left/right. Total thrust gains a sideways component.

UP / DOWN

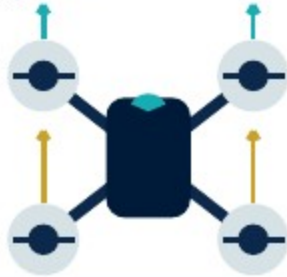
Change total thrust while keeping the body nearly level.



Angular motion comes from unequal rotor forces

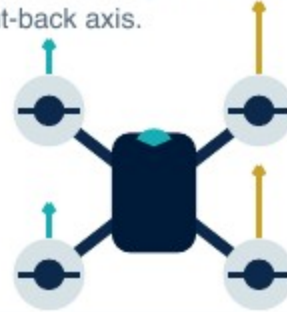
PITCH TORQUE

More thrust on one fore/aft side rotates the body about its left-right axis.



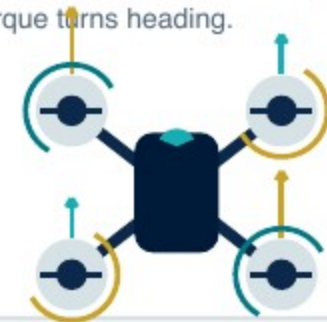
ROLL TORQUE

More thrust on one left/right side rotates the body about its front-back axis.



YAW TORQUE

Shift power between counter-rotating rotor pairs; reaction torque turns heading.



ROLL ϕ



PITCH θ



YAW ψ



Angular acceleration follows net torque, not simply motor speed.



COMMAND TO ROTOR CHANGES

Motor mixing at a glance

COMMAND	ROTOR CHANGE	RESULT
Up	All four increase together	More total thrust
Down	All four decrease together	Less total thrust
Pitch	Fore/aft pair difference	Nose tilts
Roll	Left/right pair difference	Body tilts sideways
Yaw	One spin pair up; opposite pair down	Heading rotates

The flight controller performs this mixing continuously; student code requests motion, not raw motor RPM.

Roll, pitch, and yaw rotate about different body axes

ROLL · FORWARD BODY AXIS

ϕ about x_b

PITCH · LATERAL BODY AXIS

θ about y_b

YAW · VERTICAL BODY AXIS

ψ about z_b

TORQUE → ANGULAR ACCELERATION

$$\alpha = \mathbf{I}^{-1} \tau$$

Body axes: x_b points forward, y_b points laterally, and z_b is vertical. The sign of z_b depends on the chosen convention.



From code to stabilized motion



```
HOPPER PROGRAM.JS

1  await drone.takeOff();
2  await drone.fly("forward", 1, 15);
3  await drone.hover();
4  await drone.land();
```

WHY SEQUENCING MATTERS

- await keeps commands in order.
- Timed fly and rotate commands are open-loop motion requests.
- Built-in settle time reduces overlap between maneuvers.
- Vision checks should happen after the aircraft stabilizes.



Teach the model, then test the model

QUICK DEMONSTRATIONS

- Hold a paper quadrotor level: where does thrust point?
- Tilt it 10 deg: what horizontal force appears?
- Compare all-motor change with paired-motor change.
- Predict motion before running a 15% simulator command.

SOURCES AND FURTHER READING

NASA STEM - The Science Behind Quadcopters

<https://www.nasa.gov/stem-content/the-science-behind-quadcopters/>

Stanford - Quadrotor Helicopter Flight Dynamics and Control

https://ai.stanford.edu/~gabe/papers/Quadrotor_Dynamics_GNC07.pdf

Hopper Studio - classroom simulator dynamics

lib/simulation.ts in the Hopper Studio project

PREDICT -> COMMAND -> OBSERVE -> EXPLAIN

The simple equations are a reasoning tool, not a precision flight model.